

Homework on Variables, and Data types

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Fill in the Blanks

1. In Java, a variable is the name of a reserved _____ allocated in memory.
2. A variable declared inside the body of a method is called a _____ variable.
3. Instance variables are declared inside the class but outside the _____.
4. A variable declared as _____ is shared among all objects of the class.
5. The this keyword is used to refer to the current _____.
6. The size of int in Java is _____ bytes.
7. The range of byte is from _____ to _____.
8. The primitive data type that stores true/false values is _____.
9. Non-primitive data types include Classes, Interfaces, Strings, and _____.
10. In Java, variables must be _____ before they are used.

☒ True/False

1. Local variables can be declared with the static keyword.
2. Instance variables are shared among all objects of a class.
3. Static variables are initialized only once when the class is loaded.
4. The char data type in Java takes 1 byte of memory.
5. Java is a pure object-oriented programming language.

6. The this keyword can be used to differentiate between local and instance variables.
7. Primitive data types in Java include boolean, int, float, and String.
8. The long data type in Java is 4 bytes.
9. A constructor in Java has the same name as the class and no return type.
10. Java variables can store multiple values at the same time.

🔗 Multiple Choice Questions (MCQs)

1. Which of the following is NOT a primitive data type in Java? a) int b) float c) String d) char
2. What is the size of a short in Java?
a) 1 byte b) 2 bytes c) 4 bytes d) 8 bytes
3. Which keyword is used to refer to the current object?
a) super b) this c) static d) final
4. Which variable type is shared among all objects?
a) Local variable b) Instance variable c) Static variable d) None of the above
5. Which of the following is a floating-point data type?
a) int b) float c) char d) boolean
6. The range of int in Java is:
a) -128 to 127 b) -32768 to 32767 c) -2147483648 to 2147483647
d) -9223372036854775808 to 9223372036854775807
7. Which of the following is true about constructors?
a) They have a return type.
b) They must have the same name as the class.
c) They can be declared static. d) They are called manually by the programmer.
8. Which of the following is NOT a non-primitive data type?
a) Arrays b) Interfaces c) Enums d) int

9. Which of the following data types supports 15 decimal places?
a) float b) double c) int d) long
10. Which company originally developed Java? a) Microsoft b) Sun Microsystems c) Oracle d) Google

Assertion-Reason Questions

Your answer will be like A(assertion) and R(reason):

A and R are true, and R is the correct explanation.

1. **Assertion (A):** Local variables cannot be declared as static.
Reason (R): Static variables are shared among all objects of a class.
2. **Assertion (A):** Instance variables are object-specific.
Reason (R): Each object has its own copy of instance variables.
3. **Assertion (A):** Static variables are initialized only once.
Reason (R): They are created when the class is loaded into memory.
4. **Assertion (A):** The this keyword refers to the current object.
Reason (R): It is used to differentiate between local and instance variables.
5. **Assertion (A):** Java is a strongly typed language.
Reason (R): Each variable must be declared with a data type before use.
6. **Assertion (A):** The char data type in Java is 1 byte.
Reason (R): It stores ASCII values.
7. **Assertion (A):** Constructors in Java can have a return type.
Reason (R): They are special methods used to initialize objects.
8. **Assertion (A):** The double data type supports 15 decimal places.
Reason (R): It uses 8 bytes of memory.
9. **Assertion (A):** Non-primitive data types are user-defined.
Reason (R): Examples include Classes, Interfaces, and Arrays.
10. **Assertion (A):** Java variables can store multiple values at once.
Reason (R): Arrays are used to store multiple values.